

and costs of IoT deployments. Present-day technologies work largely in silos. However, in the future, a full-functional digital ecosystem will require seamless data sharing between machines and other physical systems from different manufacturers.

Moving forwards

Although it is perceived that value-added services such as applications and system integrations will be the significant revenue driver, hardware is also likely to generate sizeable revenue through its sheer volume of deployment. This would no doubt impact the demand for electronic components. Individual IoT devices will inevitably require a microcontroller (MCU) to add intelligence to the device, one or more sensors to allow for data collection, one or more chips for connectivity and data transmission, and a memory component. Connected devices that transmit information across relevant networks rely on innovation from semiconductor players—for instance, highly-integrated microchip designs and very-low-power functions.

A new array of sensors based on microelectromechanical systems (MEMS) technology is rapidly being developed to enable IoT applications beyond motion and image sensing. These would include applications that measure humidity, altitude, food calorie composition and various human health indicators.

With connectivity being a key capability required in IoT applications, multiple communication standards and protocols are being supported by consumer electronics and industrial devices, like low-power and personal area wireless mesh networks, which

TABLE III IoT APPLICATIONS IN PUBLIC SECTOR		
Category	Sub-category	Example
Transportation	Public transportation	- Using GPS tracking devices, we can enable real-time monitoring of buses and give better information on waiting times - Using swipe-card information, we can analyse peaks in use and provide a better supply of buses
	Traffic	Using GPS tracking devices or presence detection sensors, we can enable real-time traffic analytics and have a smarter control of traffic lights to prevent traffic congestion
	Public bikes	Using GPS tracking devices, we can analyse where public bikes are most needed and also balance the availability of bikes at different locations
Economy	Tourism	Using beacons, we can push notifications on smartphones based on location and give people additional information about a place in a city or an art object in a museum
	Industry	Smart infrastructures
Health	Assistance to people	Using a single-push button, we can send an emergency alert
	Automated medication	Using medical devices, we can automate the supply of medicine for people with diabetes or high blood pressure and send alerts to the doctor when the device detects a problem
	Maintenance of medical devices and failure detection	Using a sensor, a battery-powered medical device could send a notification when a battery needs to be replaced
Environment	Water supply	Using sensors, leakage in water pipes can be detected
	Pollution	Using sensors, we can monitor pollution levels and the amount a polluting particle emitted
Administration	Electronic identity device	- Using magic bands, we can have all our identity information and important information updated - Using biometrics, personal magic bands can be secured and prevent identity theft
Security	Drones	Using computer vision, drones can identify security issues and report them
	Fire	- Using smoke detectors, automated notifications can be sent to intervention centres - Using sensors in gas pipes, we can identify potential sources of fire and deal with them before any damage is caused

Source: Deloitte, 2015

are primarily suited for sensors. IoT applications usually require a single chip in an acceptable form factor and low power consumption for battery-operated devices. This requirement is a factor for growth in SoCs with

embedded cores, GPUs and integrated wireless connectivity in a single package. Progress in these technologies has allowed the development of such IoT devices as sensors that have compute, storage and network capabilities built into extremely small form factors, yet with low energy requirements.

India should look to build capabilities across areas such as sensors (to adapt to rugged climate/terrains in India) along with network infrastructure, standards and so on. Accelerators and incubators in India are already enabling startups to build innovative IoT solutions. Addressing current talent gaps in terms of cross-functional as well as specialised skillsets is imperative for the growth of the IoT in India. **EFY**



Fig. 5: Key challenges impeding IoT growth and adoption (Source: Deloitte)